Grouping Expected Fashion Feelings

ABSTRACT

The increasing presence of online shopping for clothing has brought an acute awareness to the correlation between event type and article of clothing. Recommendation systems can be guided by user history to recommend products according to the user's general shopping situation (ex. recommending date appropriate outfits to a user that uses a platform exclusively to buy date clothes).

In this project, we develop a model to predict the use case of clothes based on various features. We first conducted analysis to investigate the properties of the data to identify key features. Based on these relevant features, we developed various models to tackle the problem and selected the strongest one which employed BERT, a cutting edge NLP algorithm.

1. INTRODUCTION

The world of clothing rentals spans far and wide, with so many shapes, sizes, and reasons to rent. Our chosen dataset contains a few hundred thousand tuples, each being a sale of some clothing rental. This dataset was chosen from the collection of datasets provided by Professor McAuley. As a group, we collectively thought that it would be interesting to attempt to predict what event each rental was for, of the eight possible event type choices. These include weddings, formal affairs, parties, everyday purposes, work, dates, vacations, and other. This task was chosen for simplicity, as other categories with larger amounts of unique outcomes in the dataset would likely produce far worse accuracies. Finding success with this prediction task involved putting other attributes in the dataset to good use, such as the clothing type and review text, as these intuitively should be predictive of the event type that a piece of clothing is being rented for. The dataset also contains information about the buyer, including weight, height, body type, bust size, and many others. Furthermore, there are attributes included describing the date of purchase, rating of the item, and a summary of the review left by the user. Each attribute's predictability in relation to event type will be explored in the EDA section. In terms of how the model's performance will be measured, we decided to use the balanced accuracy score. It was determined that this would be a suitable metric since it allows us to see how accurate the model is at predicting each of the categories equally, rather than favoring certain categories which are overrepresented.

2. EDA

The runway dataset contains a near-even mix of numerical and categorical attributes, allowing for many different methods of exploration. The notable numerical attributes include weight, height, rating, age, and size. Notable categorical attributes include bust size, rented for, body type, category, year, and month. There are also two textual attributes; review text and review summary. Since handling of missing data was not covered in the scope of this course, tuples with missing data in any column were dropped. After dropping the missing data, 146,137 tuples remained in the dataset. The first step in the EDA process on this dataset was to determine the label outcome. We initially assumed that the category column would be best to use, but with so many outcomes (68 different categories), this would prove to be an extremely difficult prediction task. Upon further exploration, we found that the "rented for" attribute would be much more appropriate to use as a label outcome, since there are only 8 event type outcomes, with each containing a substantial amount of samples.

To begin our exploration of which attributes should be used in our model, it is first worth mentioning that little could be done in terms of EDA on the textual attributes, as it was assumed that NLP techniques such as word2vec would be used in model development on these. To preface how all other attributes were approached in EDA, plots were first constructed for each to visualize whether it seemed as if the distributions for each event type looked similar to one another. Box plots, line plots, and histograms were used for numerical attributes, and bar plots were used exclusively for categorical attributes.

In exploring whether the first numerical feature (age) was predictive of "rented for", a Pearson chi-square test was initially used, which would determine if any of the average ages per event type values were significantly different than their expected value. In this case, it only made sense for

the expected value to be the average age across all event types. After running the test, it was quickly determined that using Pearson's chi-square test would not be the best choice since it only shows significance when either one group is *far* from its expected value or multiple groups are offset enough from their expected values. It was then decided that a t-test, which in this case would compare whether each event type's age distribution is significantly different from the concatenated distribution of ages for all other event types, might be more appropriate to use. More specifically, a Welch's t-test is the appropriate test to use for all numerical attributes in this dataset, since there is no assumption that the two distributions have equal variances. Each numerical attribute, except for rating, had t-tests conducted for each event type against all others. Most surprisingly, when viewing the boxplots for each attribute, the distributions all looked similar enough to expect that most t-tests would be insignificant. However, the opposite occurred, with 7 of 8 of the t-tests resulting in significance (with an alpha level of 0.01) for age, 7 of 8 for size, 5 of 8 for height, and 6 of 8 for weight. For all numerical attributes, the majority of event types contained likely significantly different distributions from other event types. For this reason, all numerical attributes will be used in our final model.

EDA on the "category" column (explains the clothing type) was done using the proportions of each clothing type across event types. If for any clothing type, there were no events in which the proportion of that clothing type exceeded the designated threshold of 30%, this clothing type would be considered being changed to "other" in the category column. This would be done because for any of these clothing types, we might see a somewhat uniform distribution across event types, meaning this clothing type is not more associated with any particular event type. Therefore, all of the clothing types changed to "other" would eventually be One-Hot Encoded into the same insignificant category. Only two of the 68 clothing types had no event types with a proportion above the threshold. For both of the clothing types, the proportions were inspected. While none of their proportions lied above the threshold, "shift" appeared to be mainly contained in only 2 event types (both containing 27-28% of them), and "skirts" were evenly distributed (25% each) among 4 of the 8 event types. Since neither clothing type appeared to show a near uniform distribution across event types, neither were changed to "other", and it was determined that category could be useful in predicting event type.

For the year attribute, a line plot of count of sales for each event type over the years was plotted to visualize if rentals for certain events were more popular in certain years. This was indeed the case, which was enough to motivate use of year in model dev.

Determinations of whether categorical attributes are potentially predictive of our label outcome were done in a way similar to how numerical attributes were handled, in the sense that, for each attribute, each event type distribution was tested against a concatenated distribution of all other event types. Rather than using a t-test, though, permutation testing was used. Permutation testing determines if two distributions are significantly different. Permutation tests were used for all categorical attributes, and were also used on rating, since it is not a continuous numerical attribute. Again, when comparing the bar plots and histograms of the distributions, they seemed similar enough to almost assume that there would be no significance with any of the permutation tests. We were again surprised when most results showed the opposite. All tests were statistically significant for month, 6 of 8 tests were significant for rating, 5 of 8 significant for body type, and 7 of 8 significant for fit. Since once again the majority of event types contained likely significantly different distributions from other event types for all categorical attributes, we decided that all can be used in model development.

3. Literature Review

3.1 Data Origin

The dataset we employed in our work originates from the collection provided by Professor Mcauley. It was originally used to address the product size recommendation problem [1] where regardless of the "true" size of any piece of clothing, users may not necessarily interpret it the same when worn, resulting in a shortcoming of the recommendation system. The dataset contained values such as customer measurements and fitting feedback from RentTheRunway, an online clothing rental site and was the foundation to generate a model that accounted for minute user preferences.

3.2 Related Work

The text based classification problem is a well established one with many studies exploring the problem space and introducing new solutions to address the growing topic. Many of these investigate similar problems where labels/categories need to be derived from a large corpus. For example, a common topic of investigation is predicting ratings based on review corpus.

Some of the more common approaches include Logistic Regression, Naive Bayes, SVM's, Decision Trees, Artificial NN's, and much more [2]. Depending on the use cases, these models can yield potent results, but they all have their unique shortcomings that limit their generalizability. Modern, cutting edge solutions trend towards a more recent development in the NLP field, transformers.

State of the art ventures in the field largely employ transformers to process the textual data. One of the most advanced transformer based models in practice currently is **B**idirectional **E**ncoder **R**epresentations from **T**ransformers (**BERT**). BERT is a pre-transformer that can be fine-tuned with just a single additional output layer. With such minor work, it achieves outstanding results in common NLP tasks [3]. The performance levels that BERT can produce are marginally better than traditional NLP approaches, thwarting them in terms of accuracy in a set of common textual based tasks [4]. The general consensus as of now is that BERT is the optimal solution to problems that require NLP work to be done (i.e. the presence of large text data).

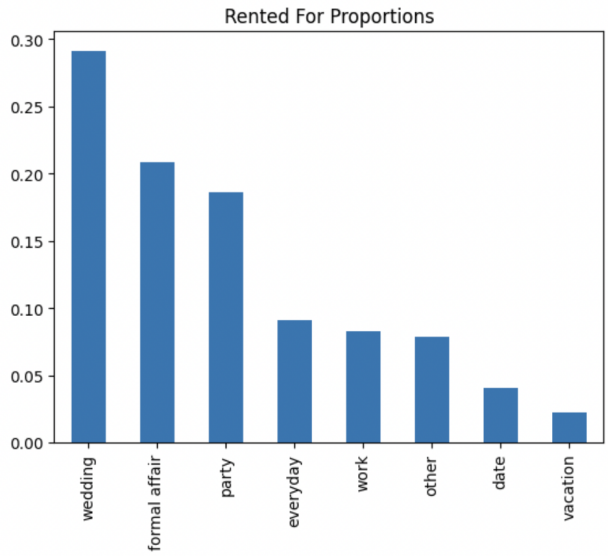
Following these findings, we decided to explore multiple models and approaches, but eventually landed on BERT for our NLP as it had by far the best performance. This lines up with the conclusions drawn from other researchers where BERT based results are able to achieve a significantly better accuracy than their counterparts.

4. MODELING

4.1 Baseline Model

4.1.1 Intuition Behind Model Choices

Prior to creating the chosen baseline model, we considered a variety of options given the task of category prediction. With 8 possible classes, an initial starting point of simply randomly choosing one of the eight classes would result in a 12.5% accuracy rate, assuming the distribution of rental occasions was uniform. However, such a baseline faces a couple of issues. Firstly is the fact that such a model is not deterministic. That is to say, predictions resulting from model use are different with each model run, making the predictions not informative of any relationship in our data. Secondly is the fact that there is an imbalance in the representation of the classes in our data (refer to image)



This leads to the second possible baseline which is just to predict the most popular class. Given the class distribution of our data, if the sample is representative of the population that when only predicting the most popular class we can expect to get it correct 29.13% of the time. However, one restriction of this model is that it would logically only predict one of the categories correctly while being incorrect for all other categories 100% of the time. This leads to a balanced accuracy score of 12.5% which is roughly similar to that of the random choice baseline.

Because the previously mentioned models have fundamental failings, they would prove to be poor comparisons to the final models we create. With this in mind, we sought a more robust baseline model that takes into consideration some aspects of the data in its decision. Taking inspiration from a similar task done in Assignment 1 of our CSE 158 class, we chose to use a baseline model that simply predicts the occasion based on the counts of the occasion's title within the review, predicting the most common

title (or "other" if no category title is present). While not taking advantage of the full dataset, this still would still provide interesting results in regards to how user's reviews inform the typical occasion an outfit would be used for.

4.1.2 Results

Our resulting baseline implementation garnered a balanced accuracy score of 26.4%. Notably, a large number of reviews in the dataset made no explicit mention of their category, resulting in a significant overprediction of the "other category". Furthermore, the more populated categories, such as "wedding", "formal affair", or "party" were also severely under-guessed, despite an intuitively high likelihood of unique wear specific to these types of occasions. Despite this, occasions with explicit dress codes or attire requirements were still guessed correctly a large number of times, in part to the high-likelihood of explicit mention in user reviews. More everyday-type attire choices were not as likely to be spelled out in review text, leading to a low correct percentage for categories like "everyday" or "vacation".

4.2 Final Model

4.2.1 Intuition Behind Feature Creation

The first step in our modeling process was of course creating features out of the columns we have in our dataset. The most simple of these were the quantitative variables which we chose to be *weight, height, size,* and *rating*. To turn these variables into features we chose to standardize them to ensure that the units for our features are equal (a formula could be added). This was necessary during model development if we wanted to use regularization terms.

Next, we took a look at our only ordinal variable which included *bust size*. Because it is an ordinal variable, we turned it into a feature by establishing its order (a-j) and encoding it as corresponding integers (0-12). This can then be treated like our other numerical variables for standardization and for use in our model.

Following were our nominal non-text-based variables which we chose to be *body type, age, year, and month*. Because our decision to consider these as nominal may be questioned, the intuition behind some of these choices will be given. With *age*, while there is a clear numerical component, using it as a quantitative variable implies a relationship as years increment by one which makes little sense for most applications, including our own task where our EDA shows little correlation when looking at age year to year but a much higher correlation when looking at age as different groups (youth, adult, elderly, etc.). With year and month, we chose to treat these as categorical because changes in how fashion choices are generally known tend to change based on the season or are defined by yearly general trends. This is in contrast to some gradual change which would be the assumption if we treated these variables as quantitative variables.

With this intuition in mind, we chose to turn our nominal variables into columns through one-hot encoding. This includes also dropping one of the columns due to the redundant encoding that comes from not doing so.

The last variables we turned into usable features were the text-based variables, *review_text,* and *category*. There are various different methods with which to turn these variables with the most standard ways being to use a bag-of-words approach and then calculate the count or TF-IDF for each word in a given sentence. While these would be helpful in representing our text as features for a model, there is the downside that this encoding does not show the relationship between words (something that is crucial for understanding the underlying meaning behind certain words).

Acknowledging the downsides of the bag of words approach we chose instead to use an existing transformer model and create sentence embeddings through it. More specifically we used the *sentence-transformers* tool provided by Hugging Face with the *all-MiniLM-L6-v2* model[1]. This creates a 384-dimensional embedding vector that contains latent information behind the given review or category. This creates an additional 768 features for our model.

4.2.2 Intuition Behind Model Choices

After generating the features for our final model, we ended up with a total of 806 features, most of which were sparse. With this many features, it is very likely that not all of them are actually important which could lead to issues of overfitting if not handled. Because of this, we chose to just use a Logistic Regression model where we would experiment with

[1]https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

regularization terms to diminish the impact of unimportant features. This led to the creation of 3 models, one where just the standardized features were passed with no regularization, one where an L2 regularization term was added, and another with L1 regularization term.

For the L2 loss model, recall that the formula for this regularization term that is added to the loss is as follows:

$$\lambda \sum_{i=1}^{N} w_i^2$$

Where N is the number of features, w_i is the weight for a given feature i and λ is the strength of the regularization term. The decision for what strength to use was made using stratified 5-folds cross-validation on different penalty terms (values from .0001 to 1000 by orders of magnitude). This gave the result of .1 for λ.

L1 loss has a similar formula except that it uses the absolute value of the weights instead of squared values.

$$\lambda \sum_{i=1}^{N} \left| w_i \right|$$

This has a tendency to make weights equal to zero for certain features when optimizing which acts as a form of feature selection. Given limited computation power for solving with an L1 penalty term, we only performed cross-validation on .1, 1, and 10 for the values of λ which resulted in the optimal value of .1 for λ.

4.2.3 Results

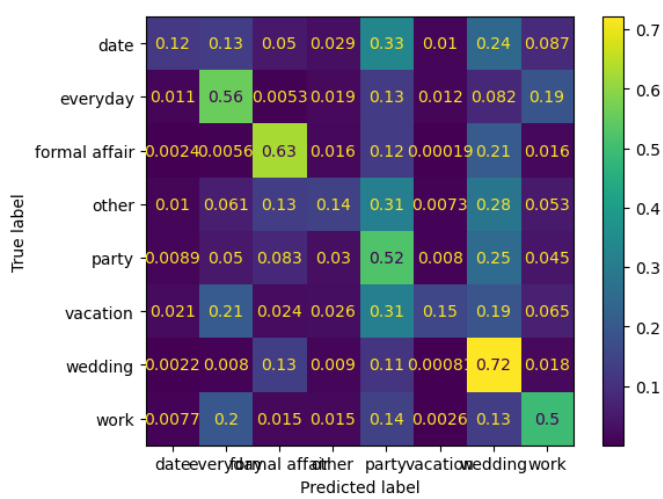
With the three models we trained them using a training set of 65% of our dataset and then calculated the balance accuracy score of each model for assessment:

Penalty Type	Balance Accuracy Score
None	0.41780
L1	0.41781
L2	0.41800

These results show that while there were only marginal differences in balanced accuracy between using and not using a regularization term, it does seem to have a positive effect as can be seen in the L2 loss results. This implies that our model contained features that were either redundant in its encoding or just noise when it came to predicting our result.

One aspect of these results that are worth noting is the similarities between the L1 penalty model accuracy and the no penalty model accuracy. While there is a very slight improvement by the L1 model, this result is negligible which may seem to indicate that L1 loss had very little effect on the final accuracy. However, it's once again important to note that due to hardware limitations, it was much harder to test a wider range of penalty strengths which likely impacted the results we found.

Although our balanced accuracy scores show only modest improvement over baseline models, this may be due to the nature of the predictive task at hand. The semantics of a review may not be actually indicative of a use case. Furthermore, plenty of clothing items can be expected to qualify for more general use, while niche items (such as a wedding dress) might be exclusively used for certain events. Specifically, we see that the model is incredibly effective at predicting "wedding", "everyday", and "date". In other cases, the model struggles, often picking from "everyday", "party", "wedding", and "vacation" when the clothing is from "vacation". Thus, we find that the model is adept at identifying items that are assuredly used for only special occasions, while struggling on items that might have a home in multiple categories.



5. CONCLUSION

Predicting categorical use cases for an article of clothing proved to be a difficult task, especially when trying to discern between more everyday-like occasions. Categories like “work”, “everyday”, or “date” may have loose enough dress codes to encompass a wide variety of uniform, while “formal affairs” and “weddings” may see a significantly smaller variety of attire. This distinction is made clear through the performance and limitations of our model, which show that these niche use cases are more easily predictable.

The sentence transformer models were an improvement over more simplistic baseline models. The baseline models struggled on the majority of categories, with the exception of “other” and (to a lesser extent) “work”. In contrast, the sentence-transformer based models perform with much greater accuracy on distinct categories, like “wedding”, “everyday”, “work” and “formal affair”. These struggles can be attributed to the nature of the predictive task. A type of clothing may not necessarily be confined to a single occasion or use type. In cases where a particular occasion requires specific dress codes or formalities, the model saw gains in performance. Additionally, marginal improvements made using penalty terms show that while there may be some redundancies in feature encodings, the majority of included information is predictive of a clothing’s use-case category.

6. REFERENCES

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